



OpenSpatialDelay

Third-Party Notices and Attributions

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spatialmedialab.org | github.com/Spatial-Media-Lab/OpenSpatialDelay

Software Libraries

OpenSpatialDelay is built with the following third-party software libraries:

Component	Version	License	Copyright Holder
JUCE (Audio Plugin Framework)	8.x	GPL-3.0 / Commercial	Raw Material Software Limited
libmysofa (SOFA File Reader)	1.3.2	BSD-3-Clause	Christian Hoene, Symonics GmbH
zlib (Compression)	1.x	zlib License	Jean-loup Gailly, Mark Adler

JUCE 8

JUCE is a cross-platform C++ framework for audio plugin development. It is dual-licensed under the GNU General Public License v3.0 and a commercial license from Raw Material Software Limited.

OpenSpatialDelay uses JUCE under the GPL-3.0 license. The full GPL-3.0 text is included in the LICENSE file distributed with the OpenSpatialDelay source code.

Website: juce.com

libmysofa v1.3.2

libmysofa is a C library for reading SOFA (Spatially Oriented Format for Acoustics) files containing HRTF measurements.

Website: github.com/hoene/libmysofa

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zlib

zlib is a general-purpose compression library used by libmysofa for reading compressed SOFA files.

Website: zlib.net

zlib License

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HRTF Datasets

OpenSpatialDelay embeds five SOFA (Spatially Oriented Format for Acoustics) datasets for Head-Related Transfer Function (HRTF) binaural rendering. Each dataset was measured from a different head/microphone and provides a distinct spatial character.

Profile	Dataset	License	Source Institution
Immersive	SADIE II D2 KU100	Apache License 2.0	University of York
Natural	CIPIC Subject 003	Public Domain	UC Davis CIPIC Lab
Precise	HUTUBS PP2	CC BY 4.0	TU Berlin
Spatial	Bernschuetz KU100 2°	CC BY 3.0	TH Köln
Studio Reference	MIT KEMAR Large Pinna	MIT License	MIT Media Lab

Note: The Simple (Low CPU) profile uses an internal Woodworth head model and does not incorporate any third-party data.

MIT KEMAR Large Pinna

HRTF measurements of a KEMAR mannequin with large pinnae, captured by the MIT Media Lab. Used as the Studio Reference profile.

Reference: sound.media.mit.edu/resources/KEMAR.html

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SADIE II D2 KU100

HRTF measurements of a Neumann KU100 dummy head captured as part of the SADIE II project at the University of York. Used as the Immersive profile.

Reference: Gavin Kearney, Tony Doyle, york.ac.uk/sadie-project/database2

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CIPIC Subject 003

HRTF measurements of a human subject from the CIPIC (Center for Image Processing and Integrated Computing) Interface Laboratory at UC Davis. Used as the Natural profile.

This dataset is in the **public domain** and may be used without restriction.

Reference: ece.ucdavis.edu/cipic

HUTUBS PP2

HRTF measurements from the HUTUBS (Head-related Utility for Binaural Synthesis) database by the Audio Communication Group at TU Berlin. Used as the Precise profile.

Reference: Fabian Brinkmann, Alexander Lindau, Stefan Weinzierl, et al. depositonce.tu-berlin.de

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Bernschütz KU100 2°

High-resolution (2° grid) HRTF measurements of a Neumann KU100 dummy head by Benjamin Bernschütz at TH Köln (Cologne University of Applied Sciences). Used as the Spatial profile.

Reference: audiogroup.web.th-koeln.de

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